

Scenario audit of Cuba's population decline, 2021–2026

Primary scenario envelope A–D: ~1.8–2.4 million of loss in 2021–2025. Illustrative central scenario (D): ≈8.59 M and about 1 in 5 on the corrected base. Against the official 2021 headcount, the gap toward 2026 approaches 1 in 4 — a different definition. Emigration accounts for ≈91% of absolute loss under D.

1.8–2.4 M

across-scenario envelope
(primary uncertainty)

8.59 M

scenario D end-2025
(not an identified estimator)

2.0×

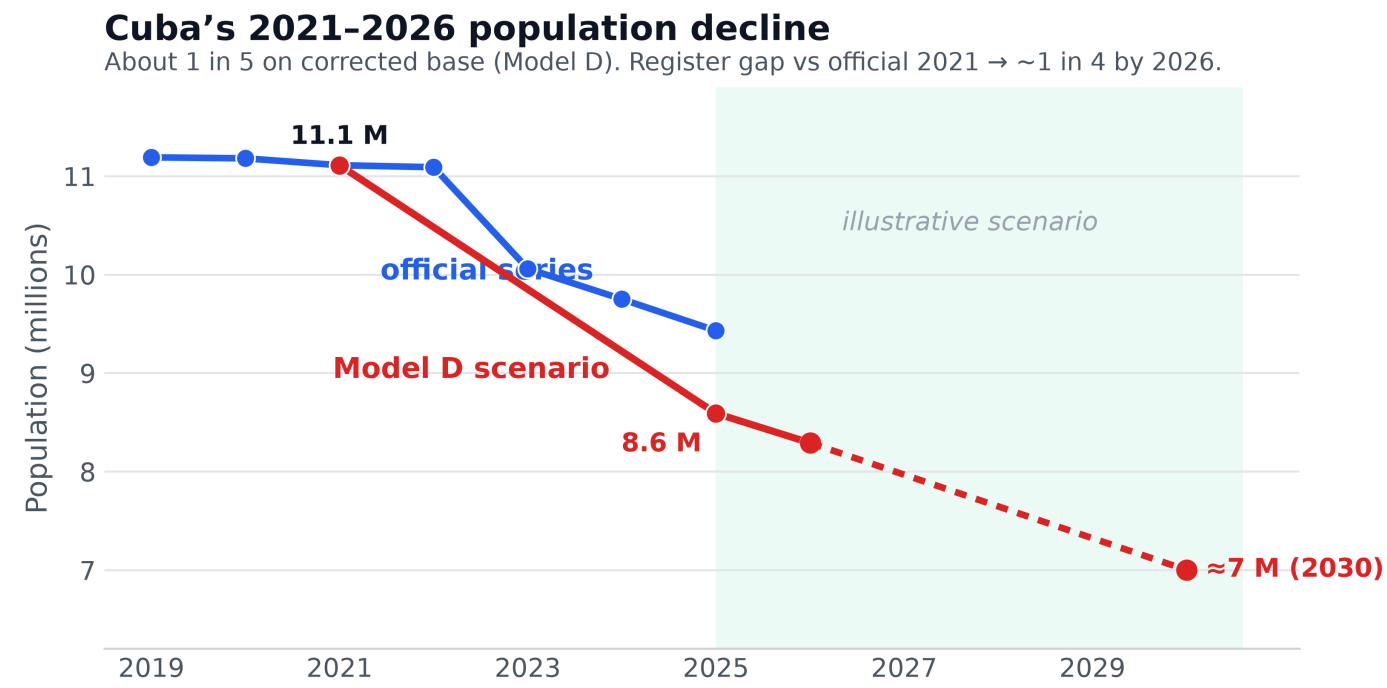
deaths roughly double births
(136k vs 68k in 2025)

1.05 M

settled-destination floor
(U.S. ~800k + non-U.S. ~250k)

Official headcount versus scenarios

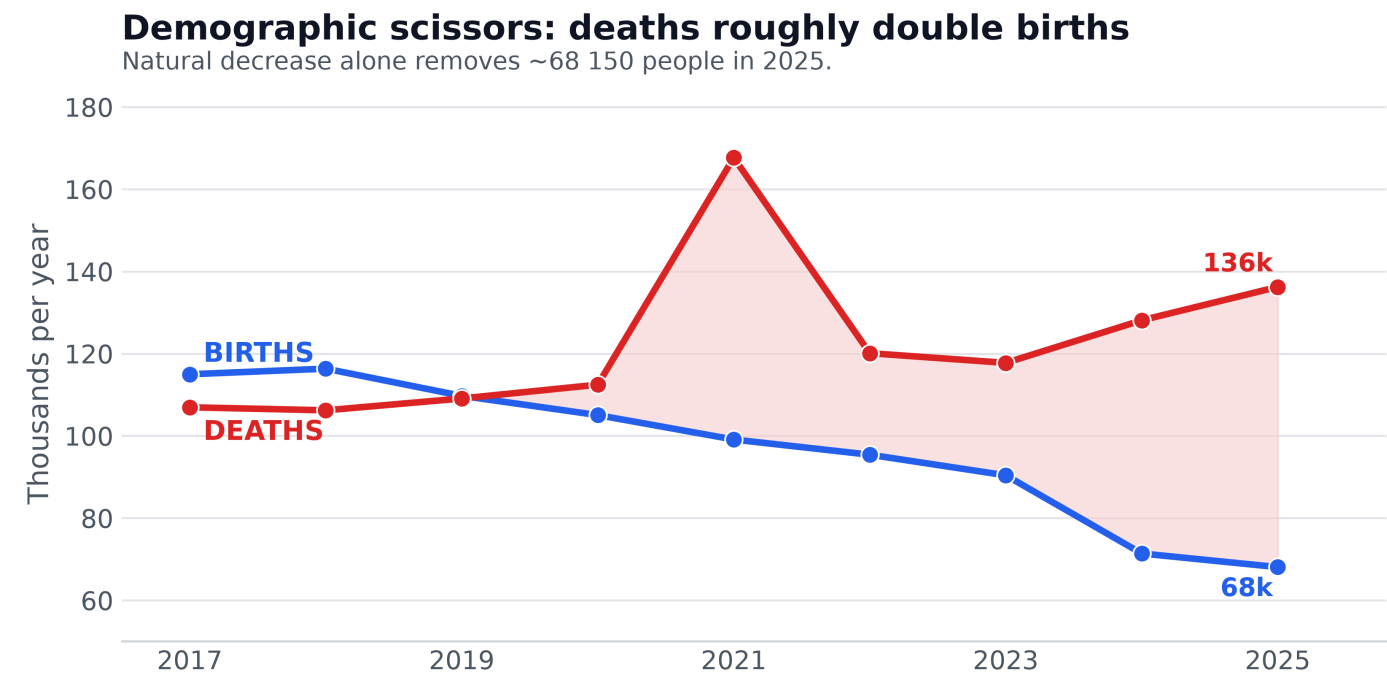
ONEI ≈9.43 M vs scenario D ≈8.59 M (end-2025). Post-2026: illustrative trend, not a forecast.



Source: ONEI / Monte Carlo scenarios (Model D = illustrative central path).

Demographic scissors: more deaths, fewer births

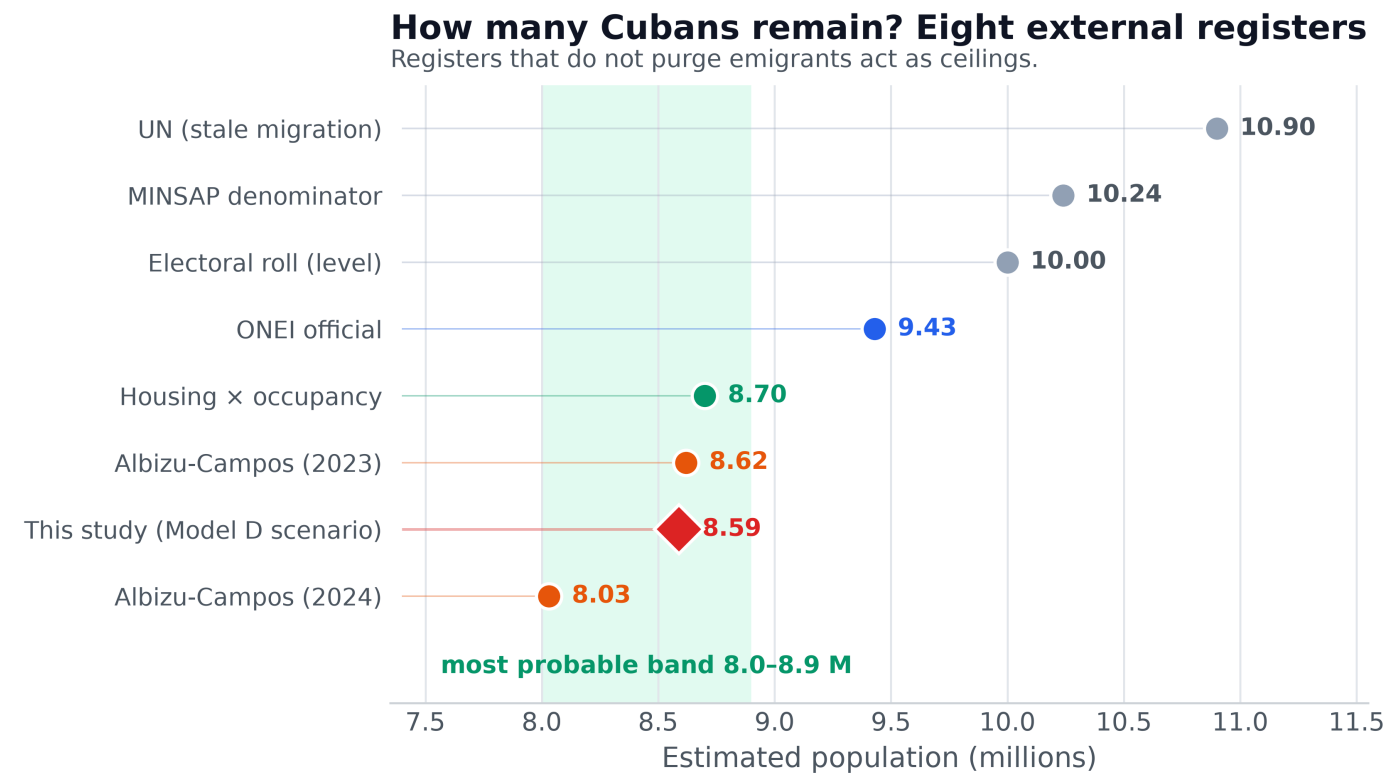
Since 2020 deaths exceed births by a wide margin. In 2025 two people die for every baby born.



Source: ONEI demographic yearbooks. The 2021 spike corresponds to the COVID wave.

Eight sources, soft band 8.0–8.9 M

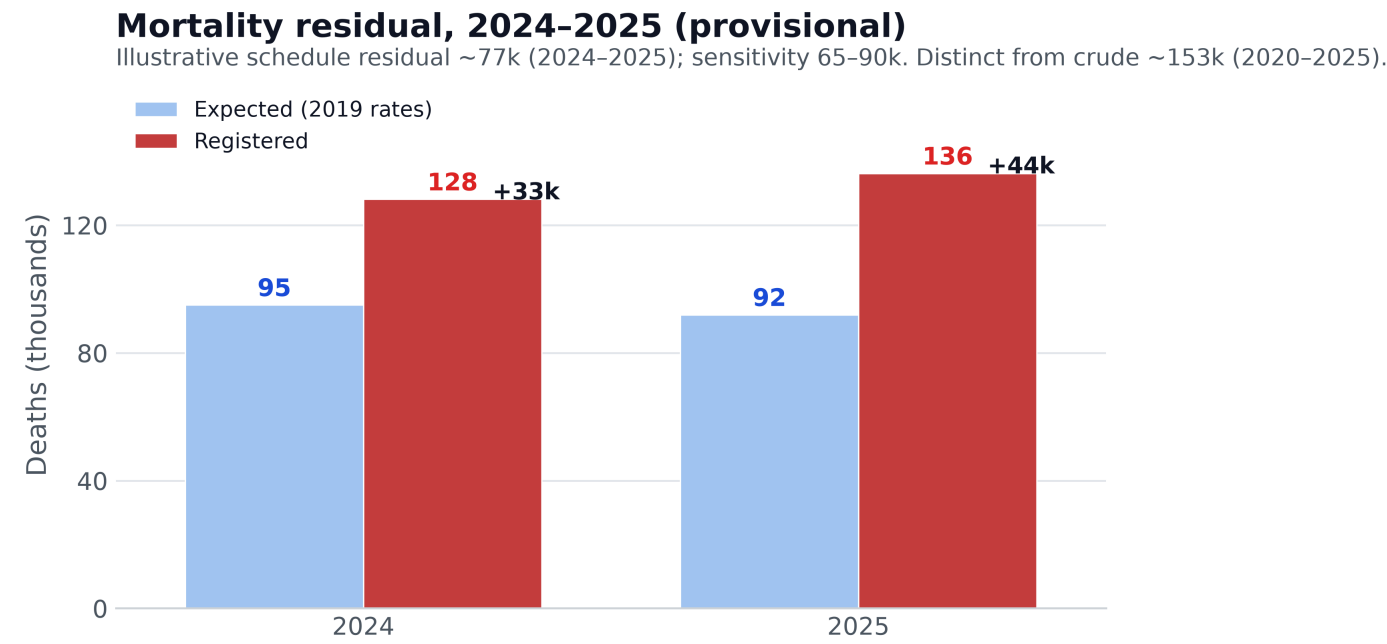
High ceilings (UN, electoral roll) vs exit-correcting anchors (Albizu, housing). Scenario D is the study estimand, not external validation.



Electoral roll, MINSAP, housing occupancy, UN, Albizu-Campos (2023, 2024), and scenario D.

Provisional mortality residual, 2024–2025

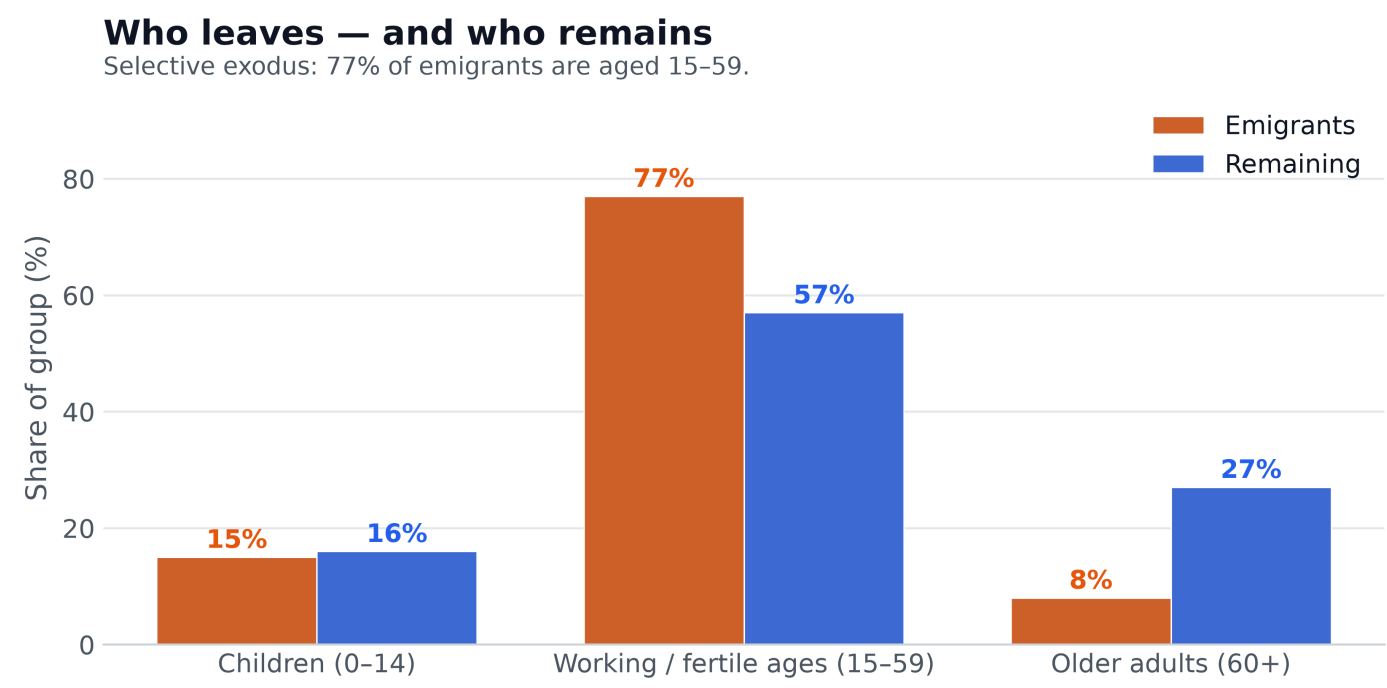
Illustrative ~77k vs 2019 schedule (band 65–90k). Distinct from crude registered excess ~153k (2020–2025). Full ONEI age–sex pipeline pending.



Not a Kitagawa decomposition; expected-deaths residual under a provisional 2019 schedule.

The young leave; the old remain

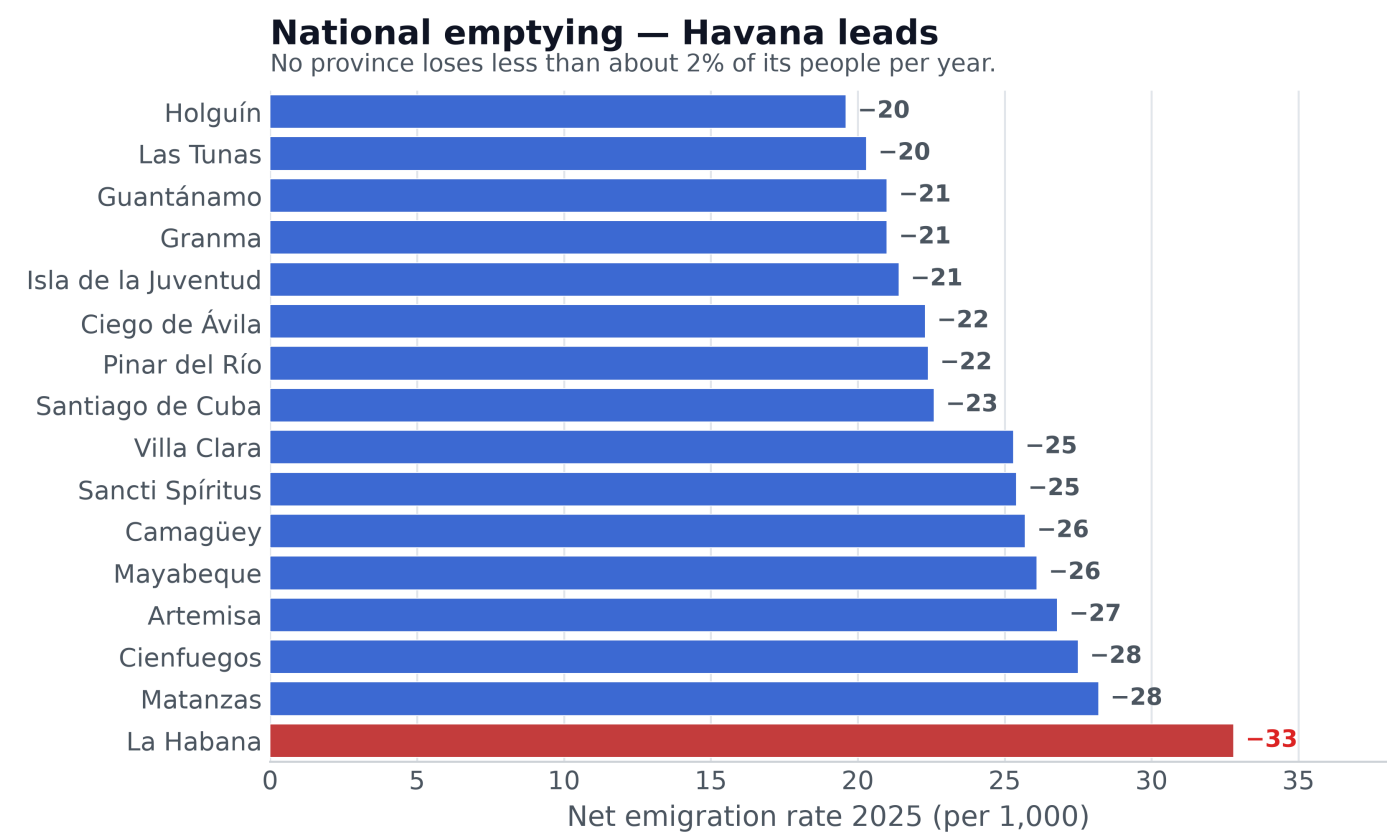
77% of emigrants are aged 15–59. Each young exit reduces future births.



Age composition of emigrants vs residents; median age leavers ≈30, stayers 45.

Emptying is national; Havana leads

All provinces lose about 2% of their people per year (ONEI; levels may be biased).



Net emigration 2025 per 1,000 inhabitants. Source: ONEI Indicators 2025.

HOW TO CITE

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METHOD

Demographic balance solved by Monte Carlo (10⁸ draws). Models A–D = scenarios; D = illustrative central path. Destination ledger floor ~1.05 M (U.S. admin + 250k non-U.S.). The step to ~2.0 M net under D is a prior on ONEI migration, not destination microdata.

SOURCES

ONEI yearbooks and Indicators 2025. UN WPP. Albizu-Campos; Brønnum-Hansen & Albizu (2023). Data and code: github.com/ypriverol/cubascience